

9. Asymptotics and Convergence

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```
#Purpose: Explore central limit theorem and the law of large numbers
#Clear environment
rm(list = ls())
library(tidyverse)
```

```
## -- Attaching core tidyverse packages ----- tidyverse 2.0.0 --
## v dplyr      1.1.4      v readr      2.1.5
## v forcats    1.0.0      v stringr   1.5.1
## v ggplot2    3.5.2      v tibble    3.3.0
## v lubridate  1.9.4      v tidyr     1.3.1
## v purrr      1.0.4
## -- Conflicts ----- tidyverse_conflicts() --
## x dplyr::filter() masks stats::filter()
## x dplyr::lag()     masks stats::lag()
## i Use the conflicted package (<http://conflicted.r-lib.org/>) to force all conflicts to become errors
```

```
library(gganimate)
set.seed(60637)
```

#In theoretical and applied research, asymptotic arguments are often made. In this section we briefly i

#What are asymptotics?

#In probability theory, asymptotic analysis is the study of limiting behavior.

#By limiting behavior, we mean the behavior of some random process as the number of observations gets l

#Why is this important?

#We rarely know the true process governing the events we see in the social world.

#It is helpful to understand how such unknown processes theoretically must behave and asymptotic theory

#----Weak Law of Large Numbers----

#What's the probability of a fair coin landing on heads? $P[H] = 0.5$

#But any individual flip will either give heads or tails.

```
sample(c('H','T'),
       size = 1)
```

```
## [1] "T"
```

```
#As we increase the number of flips, we generally get a proportion Heads close to 0.5
sample(c(1,0),
      size = 100,
      replace = T) %>%
mean()
```

```
## [1] 0.59
```

```
#This is an example of the weak law of large numbers.
```

For any draw of independent random variables with the same mean μ , the sample average after n draws, $\bar{X}_n = \frac{1}{n}(X_1 + X_2 + \dots + X_n)$, converges in probability to the expected value of X , μ as $n \rightarrow \infty$:

$$\lim_{n \rightarrow \infty} P(|\bar{X}_n - \mu| > \varepsilon) = 0$$

```
#Graphically:
n<-100

#sample coinflips
coinflip<-data.frame(heads = sample(c(1,0),
                                   size = n,
                                   replace = T),
                     x = 1:n)%>%
#cumulative heads
mutate(prop_heads = cumsum(heads) / seq_along(heads))

p <- ggplot(data = coinflip, aes(x = x, y = prop_heads)) +
  #Proportion heads
  geom_line(color = "red", size = 1.2) +
  # True probability line
  geom_hline(yintercept = 0.5, color = "blue", linetype = "dashed", size = 1) +
  # Current point
  geom_point(color = "red", size = 2) +
  # Set axis limits
  ylim(0, 1) +
  xlim(1, n) +
  # Labels and title
  labs(
    title = 'Coin Flips Converging to True Probability',
    x = 'Number of Flips',
    y = 'Proportion of Heads',
    caption = 'Red line: Sample proportion | Blue dashed: True probability (0.5)'
  ) +
  # Styling
  theme_minimal() +
  theme(
    plot.title = element_text(size = 16, hjust = 0.5, face = "bold"),
    plot.subtitle = element_text(size = 12, hjust = 0.5),
    plot.caption = element_text(size = 10),
    axis.text = element_text(size = 10),
    axis.title = element_text(size = 12)
  ) +
```

```
## Warning: Using 'size' aesthetic for lines was deprecated in ggplot2 3.4.0.
## i Please use 'linewidth' instead.
## This warning is displayed once every 8 hours.
## Call 'lifecycle::last_lifecycle_warnings()' to see where this warning was
## generated.
```

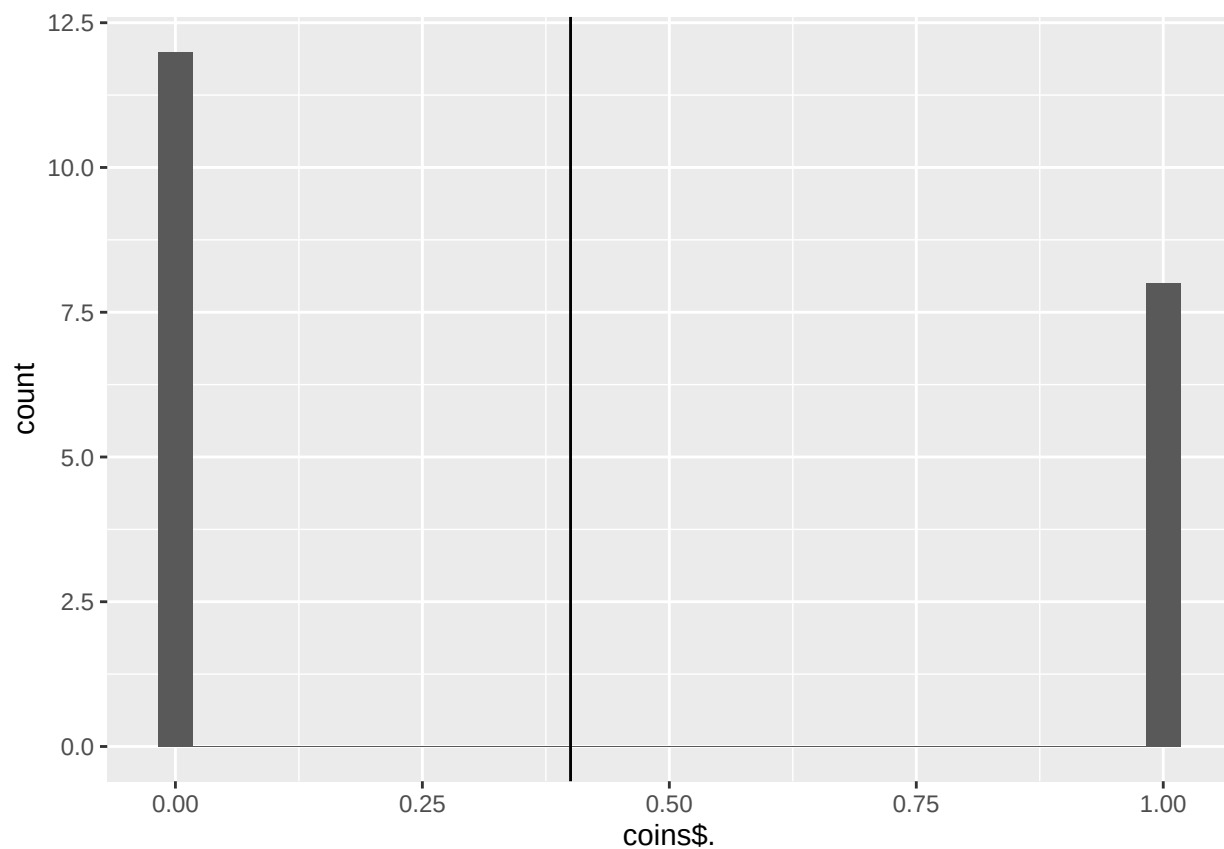
Let $\{X_n\} = \{X_1, X_2, \dots\}$ be a sequence of i.i.d. random variables with finite mean (μ) and variance (σ^2). Then, the sample mean $\bar{X}_n = \frac{X_1 + X_2 + \dots + X_n}{n}$ increasingly converges into a Normal distribution. In essence, take draws from most distributions, average them. Do that many times and those averages will approach normality.

```
#flip 20 coins (weighted)
coins<-rbinom(n = 20, size = 1, prob = .3)%>%
  as.data.frame()

#average heads
mean<-mean(coins$.)

#plot distribution
ggplot()+
  geom_histogram(aes(x = coins$.))+
  geom_vline(xintercept = mean)
```

3



```
#...NOT normal

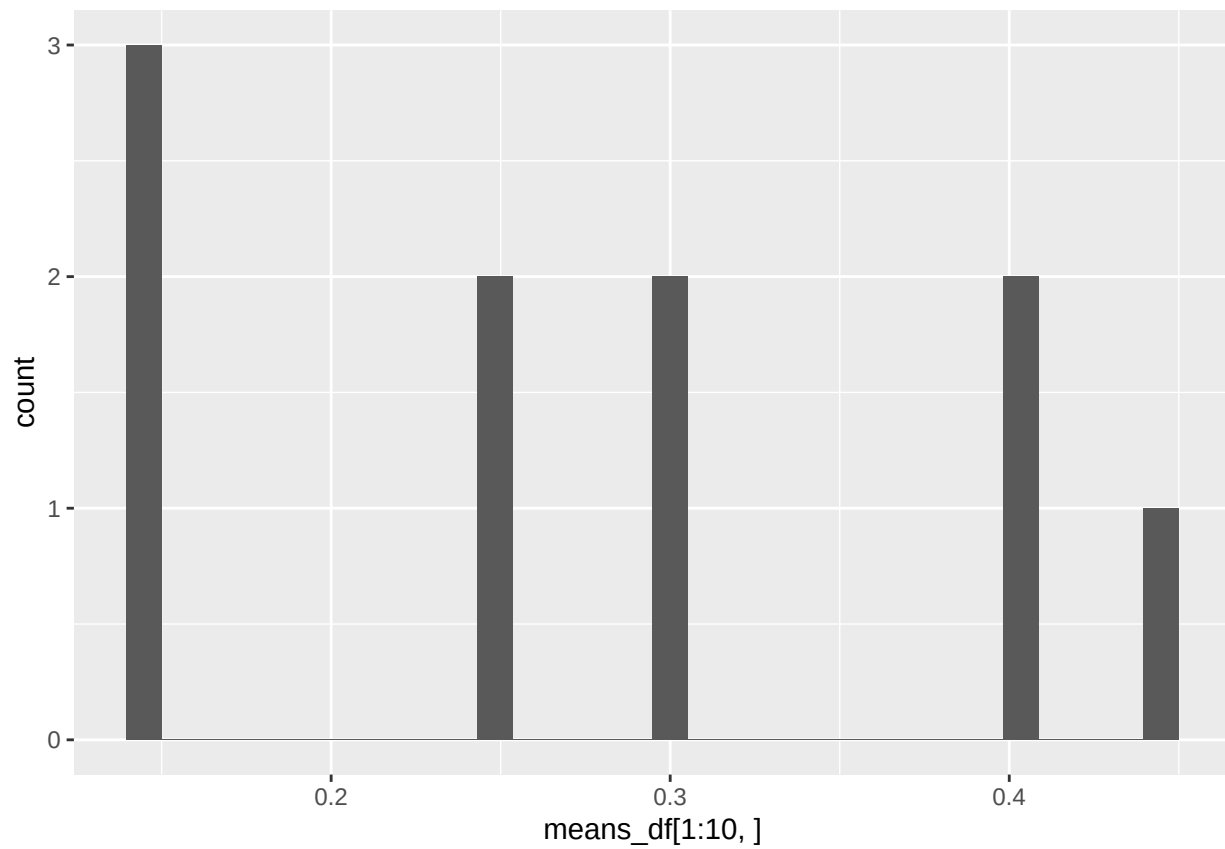
#initialize
n<-10000
mean<-rep(NA,n)

#loop
for (i in 1:n) {
  #flip 20 coins
  coins<-rbinom(n = 20, size = 1, prob = .3)%>%
    as.data.frame()
  #store mean
  mean[i]<-mean(coins$.)
}

means_df<-data.frame(mean = mean)

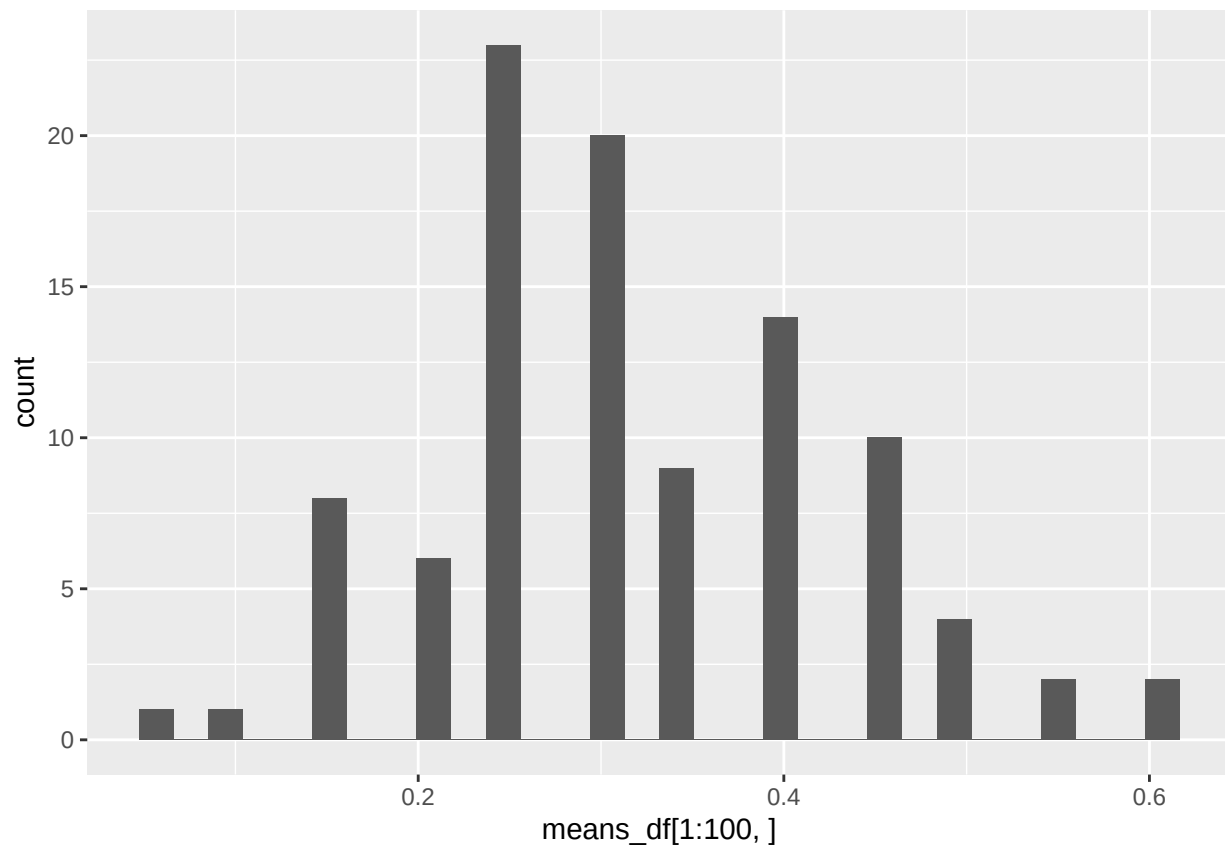
#the first 10...
ggplot()+
  geom_histogram(aes(x = means_df[1:10,]))
```

```
## 'stat_bin()' using 'bins = 30'. Pick better value with 'binwidth'.
```



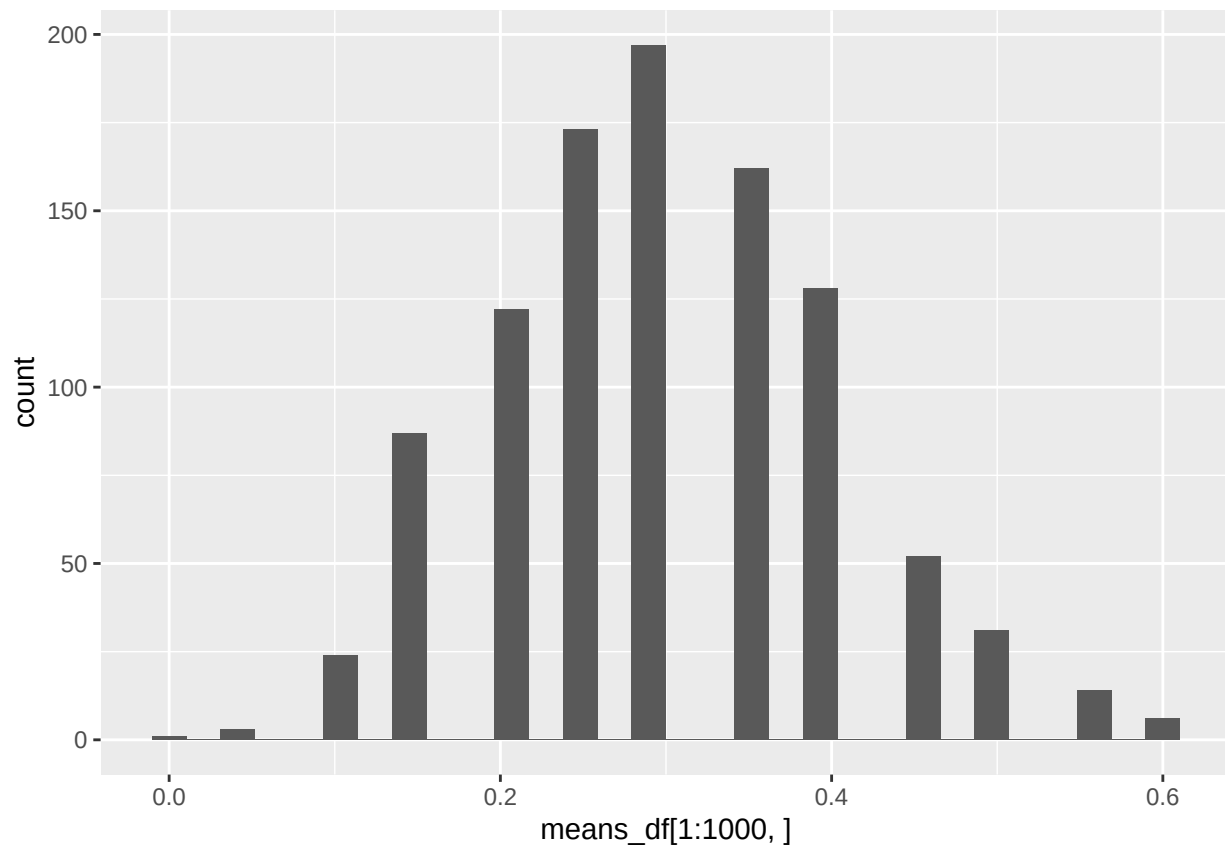
```
#...100...  
ggplot()+  
  geom_histogram(aes(x = means_df[1:100,]))
```

```
## 'stat_bin()' using 'bins = 30'. Pick better value with 'binwidth'.
```



```
#...1000  
ggplot()+  
  geom_histogram(aes(x = means_df[1:1000,]))
```

'stat_bin()' using 'bins = 30'. Pick better value with 'binwidth'.



*#This will save us in many (Not all!) cases, allowing us to assume *asymptotic* normality.*

#Exercise 1: Find a distribution from which you can sample, take means, and those means don't distribut
#Does this match any real-world situation?